

MICHAEL NJOROGÉ

SENIOR SOFTWARE ENGINEER | BACKEND & FULL-STACK | JAVA / SPRING BOOT | LARAVEL | REACT

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience designing, developing, securing and operating production software across fintech, logistics, education, ERP and e-commerce. Strong in backend architecture, REST APIs, multi-tenant platforms, databases, payments, authentication, cloud infrastructure, automated testing and production optimisation. Comfortable owning products end-to-end from architecture and implementation through deployment, troubleshooting and continuous improvement.

CORE TECHNOLOGIES

Languages: Java, PHP, TypeScript, JavaScript, Python, SQL, HTML, CSS

Backend: Spring Boot, Spring Framework, Laravel, Node.js, REST API design, multi-tenancy, microservices

Frontend: React, Vue 3, TypeScript, Vite, Tailwind CSS, Inertia.js

Data: MySQL, PostgreSQL, MongoDB, Redis, Valkey

Cloud & DevOps: Docker, Kubernetes, DigitalOcean, Hetzner, OVH, Helm, ArgoCD, GitOps, GitHub Actions, GitLab CI

Security & integrations: JWT, LDAP, API security, M-Pesa, Paystack, KRA eTIMS/DigiTax, SMS, WhatsApp, Mapbox, WebSockets, AI APIs

PROFESSIONAL EXPERIENCE

Senior Full-Stack Engineer | TerraSoftworks Limited

Nairobi, Kenya | Aug 2024 - Present

- Lead engineering on TerraGo, a production school transport and student-accountability platform built with Laravel, React/TypeScript, Redis and Kubernetes.
- Designed and implemented REST APIs and multi-tenant workflows for schools, staff, students, transport operations and parent-facing features.
- Implemented school-scoped synchronisation and reconciliation workflows for WaaS integrations, replacing inefficient parent-level processing.
- Built real-time Mapbox functionality including GPS drift correction, speed filtering, road-aligned vehicle rotation and trip replay.
- Designed SMS/WhatsApp notification workflows with per-school templates, short links, webhooks and delivery reconciliation.
- Led a V2 API cleanup and a frontend/backend audit that identified and documented 22 integration defects.

Full-Stack Engineer | Bitz IT Consulting Limited

Nairobi, Kenya | Jan 2023 - Jul 2024

- Developed and maintained RESTful APIs using Laravel/PHP for enterprise applications; automated data flows using Python ETL scripts.
- Designed relational and non-relational database schemas using MySQL, PostgreSQL and MongoDB for high-volume applications.
- Implemented LDAP authentication and JWT-based access control; built and maintained CI/CD pipelines with GitHub Actions and GitLab CI.
- Exposed backend services for frontend teams and contributed to React and Vue user interfaces.

Full-Stack Developer Intern | Bitz IT Consulting Limited

Nairobi, Kenya | Sep 2022 - Dec 2022

- Developed reporting and analytics modules using Crystal Reports and database queries.
- Automated internal workflows using Python scripts, reducing manual processes by 20%.

Systems & Infrastructure Trainee | Institute of Advanced Technology (IAT)

Mar 2022 - Oct 2022

- Configured and optimised Linux and Windows servers supporting 100+ concurrent student laboratory environments.
- Supported networking and laboratory systems to maintain reliable operations and minimise downtime.

SELECTED PRODUCTS & ENGINEERING PROJECTS

TechMart Africa - POS, Storefront & E-commerce Platform | Founder / Lead Engineer

Laravel | Vue 3 | TypeScript | Dexie/IndexedDB | Docker | Hetzner

- Sole architect, developer and operator of a multi-tenant SaaS POS and e-commerce product with a customer-facing storefront.
- Designed an offline-first PWA using service-worker caching, IndexedDB/Dexie storage and an idempotent synchronisation queue with UUID-based conflict resolution.
- Built StockLedgerService as a row-locked source of truth for stock mutations across sales, transfers, adjustments, purchases and offline synchronisation.
- Audited and resolved cross-tenant data-isolation vulnerabilities; introduced automated CI checks to detect tenant-scope bypasses.
- Integrated KRA eTIMS/DigiTax, split and multi-payment checkout, Paystack billing, real-time broadcasting, AI product descriptions and HR/payroll workflows supporting historical data for 800 employees.
- Built automated Playwright coverage for authentication, POS checkout, offline sales and synchronisation alongside Laravel feature tests.

Dairy Management Platform | Product / Engineering Project

dairy-app.techmartafrica.org

- Built and operate a dedicated web product for dairy-management workflows within the TechMart Africa product portfolio.

TerraGo - School Transport & Student Accountability | Production SaaS Platform

Laravel | React/TypeScript | Kubernetes | Redis | Mapbox

- Production platform supporting live transport tracking, school operations, staff workflows, parent-facing visibility, notifications and synchronisation.

EDUCATION

Maseno University - Bachelor of Science in Information Technology, Second Class Upper Division | 2019 - 2023